

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Dichlorophenylphosphine

**CAS No** 644-97-3

**Synonyms** Phenylphosphonous dichloride

**Product Code** **L07792**

**Address** ThermoFisher Scientific Australia Pty Ltd  
5 Caribbean Drive, Scoresby  
VICTORIA 3179, Australia

**Emergency Tel.** **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

**Telephone / Fax Numbers** Tel: 1300 735 292  
Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

Substances/mixtures corrosive to metal

Category 1

#### Health hazards

Acute Oral Toxicity  
Skin Corrosion/Irritation  
Serious Eye Damage/Eye Irritation

Category 4  
Category 1 B  
Category 1

#### Environmental hazards

No hazards identified

#### Label Elements



Exclamation Mark



Corrosion

**Signal Word****Danger****Hazard Statements**

H290 - May be corrosive to metals  
H314 - Causes severe skin burns and eye damage  
H302 - Harmful if swallowed  
AUH014 - Reacts violently with water

**Precautionary Statements**

P264 - Wash face, hands and any exposed skin thoroughly after handling  
P234 - Keep only in original packaging  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P390 - Absorb spillage to prevent material damage  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P363 - Wash contaminated clothing before reuse  
P402 - Store in a dry place  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P406 - Store in corrosion resistant polypropylene container with a resistant inliner  
P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component                    | CAS No   | Weight % |
|------------------------------|----------|----------|
| Phenyl phosphorus dichloride | 644-97-3 | >95      |

## Section 4 - First Aid Measures

**Inhalation**

If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical

|                                            |                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | attention is required.                                                                                                                                                                                                                                                                                                                                                               |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                                                                           |
| <b>General Advice</b>                      | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                                    |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                                                                     |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                                                                                 |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically. Symptoms may be delayed.                                                                                                                                                                                                                                                                                                                                      |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

### Extinguishing media which must not be used for safety reasons

Contact with water liberates toxic gas. Water.

### Hazardous Decomposition Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Oxides of phosphorus, Phosphorus trihydride (phosphine), Hydrogen chloride gas.

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Reacts violently with water.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

### Environmental Precautions

Should not be released into the environment.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Do not expose spill to water.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Do not allow contact with water. Handle under an inert atmosphere.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep away from water or moist air. Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Store under an inert atmosphere. Protect from moisture.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

The product does not contain any hazardous materials with occupational exposure limits established.

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | AS/NZS 2161     | (minimum requirement) |
| Viton (R)      | recommendations   |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:**

Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ

**Recommended half mask:-** equivalent)  
Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                          |                          |
|------------------------------------------------|--------------------------|--------------------------|
| <b>Appearance</b>                              | Clear                    |                          |
| <b>Physical State</b>                          | Liquid                   |                          |
| <b>Odor</b>                                    | pungent                  |                          |
| <b>Odor Threshold</b>                          | No data available        |                          |
| <b>pH</b>                                      | No information available |                          |
| <b>Melting Point/Range</b>                     | -51 °C / -59.8 °F        |                          |
| <b>Softening Point</b>                         | No data available        |                          |
| <b>Boiling Point/Range</b>                     | 225 °C / 437 °F          | @ 760 mmHg               |
| <b>Flash Point</b>                             | 102 °C / 215.6 °F        | <b>Method -</b> Open cup |
| <b>Evaporation Rate</b>                        | No data available        |                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                   |
| <b>Explosion Limits</b>                        | No data available        |                          |
| <b>Vapor Pressure</b>                          | No data available        |                          |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)              |
| <b>Specific Gravity / Density</b>              | 1.319                    |                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                   |
| <b>Water Solubility</b>                        | Reacts with water        |                          |
| <b>Solubility in other solvents</b>            | No information available |                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                          |
| <b>Autoignition Temperature</b>                | 159 °C / 318.2 °F        |                          |
| <b>Decomposition Temperature</b>               | No data available        |                          |
| <b>Viscosity</b>                               | No data available        |                          |
| <b>Explosive Properties</b>                    | No information available |                          |
| <b>Oxidizing Properties</b>                    | No information available |                          |
| <b>Other information</b>                       |                          |                          |
| <b>Molecular Formula</b>                       | C6 H5 Cl2 P              |                          |
| <b>Molecular Weight</b>                        | 178.99                   |                          |

## Section 10 - Stability and Reactivity

|                                         |                                                                                                                                          |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactivity</b>                       | Yes                                                                                                                                      |
| <b>Stability</b>                        | Reacts violently with water. Moisture sensitive. Air sensitive.                                                                          |
| <b>Conditions to Avoid</b>              | Incompatible products, Excess heat, Exposure to moist air or water, Exposure to moisture, Exposure to air.                               |
| <b>Incompatible Materials</b>           | Strong oxidizing agents.                                                                                                                 |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Oxides of phosphorus. Phosphorus trihydride (phosphine). Hydrogen chloride gas. |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                                 |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

|            |                   |
|------------|-------------------|
| Oral       | Category 4        |
| Dermal     | No data available |
| Inhalation | No data available |

| Component                    | LD50 Oral              | LD50 Dermal | LC50 Inhalation |
|------------------------------|------------------------|-------------|-----------------|
| Phenyl phosphorus dichloride | 500-2000 mg/kg ( Rat ) |             |                 |

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

##### (d) respiratory or skin sensitization;

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; No data available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

## Section 12 - Ecological Information

**Ecotoxicity effects** Do not empty into drains. Reacts with water so no ecotoxicity data for the substance is available.

#### Persistence and Degradability

**Persistence** Soluble in water, Persistence is unlikely, based on information available.

**Degradability** Reacts with water.

**Degradation in sewage treatment plant** Water reactive.

**Bioaccumulative Potential** Bioaccumulation is unlikely Product does not bioaccumulate due to reaction with water

| Component                    | log Pow | Bioconcentration factor (BCF) |
|------------------------------|---------|-------------------------------|
| Phenyl phosphorus dichloride |         | 2 - 10 dimensionless          |

|                                        |                                                                                                                                                                                                                        |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mobility</b>                        | The product is water soluble, and may spread in water systems. Reacts with water. Will likely be mobile in the environment due to its water solubility: Is not likely mobile in the environment Highly mobile in soils |
| <b>Endocrine Disruptor Information</b> | This product does not contain any known or suspected endocrine disruptors                                                                                                                                              |
| <b>Persistent Organic Pollutant</b>    | This product does not contain any known or suspected substance                                                                                                                                                         |
| <b>Ozone Depletion Potential</b>       | This product does not contain any known or suspected substance                                                                                                                                                         |

## Section 13 - Disposal Considerations

|                                            |                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations. |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                                                                                                                            |
| <b>Other Information</b>                   | Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms.             |

## Section 14 - Transport Information

### IMDG/IMO

|                             |                             |
|-----------------------------|-----------------------------|
| <b>UN-No</b>                | UN2798                      |
| <b>Proper Shipping Name</b> | PHENYLPHOSPHORUS DICHLORIDE |
| <b>Hazard Class</b>         | 8                           |
| <b>Packing Group</b>        | II                          |

### ADG

|                             |                             |
|-----------------------------|-----------------------------|
| <b>UN-No</b>                | UN2798                      |
| <b>Proper Shipping Name</b> | PHENYLPHOSPHORUS DICHLORIDE |
| <b>Hazard Class</b>         | 8                           |
| <b>Packing Group</b>        | II                          |

| Component                                        | Hazchem Code |
|--------------------------------------------------|--------------|
| Phenyl phosphorus dichloride<br>644-97-3 ( >95 ) | 2X           |

### IATA

|                             |                             |
|-----------------------------|-----------------------------|
| <b>UN-No</b>                | UN2798                      |
| <b>Proper Shipping Name</b> | PHENYLPHOSPHORUS DICHLORIDE |
| <b>Hazard Class</b>         | 8                           |
| <b>Packing Group</b>        | II                          |

|                               |                                 |
|-------------------------------|---------------------------------|
| <b>Environmental hazards</b>  | No hazards identified           |
| <b>Special Precautions</b>    | No special precautions required |
| <b>Additional information</b> | None known                      |

## Section 15 - Regulatory Information

**Safety, health and environmental regulations/legislation specific for the substance or mixture****National Regulations****Australia**

See section 8 for national exposure control parameters.

**Standard for the Uniform Scheduling of Medicines and Poisons**

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                               | Standard for the Uniform Scheduling of Medicines and Poisons                                                                                                                                         |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phenyl phosphorus dichloride - 644-97-3 | Schedule 4 listed - with anticholinesterase activity for human therapeutic use except: when separately specified in these Schedules, or in preparations containing ≤2% of Malathion for external use |

**Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                               | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|-----------------------------------------|-------------------------------------------------------------|------------------------|
| Phenyl phosphorus dichloride - 644-97-3 | Present                                                     | -                      |

**Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

**Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory**

Not applicable

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component                    | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL       |
|------------------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|------------|
| Phenyl phosphorus dichloride | X    | X     | 211-425-8 | -      | X    | -   | X    | X     | X    | X    | X     | KE-05-0452 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations****Ozone Depletion Potential**

This product does not contain any known or suspected substance

**Persistent Organic Pollutant**

This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)**

Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**



Not applicable.

| Component                    | CAS No   | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|------------------------------|----------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Phenyl phosphorus dichloride | 644-97-3 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

Authorisation/Restrictions according to EU REACH

Not applicable

## Section 16 - Other Information

### Legend

**AICS** - Australian Inventory of Chemical Substances  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**NZS 5433:2012** - Transport of Dangerous Goods on Land  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%  
**WEL** - Workplace Exposure Limit  
**DNEL** - Derived No Effect Level  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative  
**VOC** - (Volatile Organic Compound)

**NZIoC** - New Zealand Inventory of Chemicals  
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances  
**CAS** - Chemical Abstracts Service  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
 Predicted No Effect Concentration (PNEC)  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development  
**LC50** - Lethal Concentration 50%  
**ATE** - Acute Toxicity Estimate  
**RPE** - Respiratory Protective Equipment  
**NOEC** - No Observed Effect Concentration  
**BCF** - Bioconcentration factor  
**PBT** - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>  
 Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

**Revision Date** 18-Nov-2022  
**Revision Summary** Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the

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date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**